

**CE/CZ4171**

**Internet of Things:  
Communications and Networking**

**Module 1  
Introduction to IoT**

# Part I Outline

- Module 1: Introduction to IoT
  - **A Motivating Example**
  - IoT Concept
  - IoT Trend
  - IoT Applications
  - IoT Application Enablers
  - IoT Challenges
- Module 2: General Network Architecture for IoT
  - IoT Network Architecture
  - 3-Layer Model: Functional Stack, Compute Stack
  - Cloud Computing
  - Communications with Cloud: HTTP, REST, CoAP, MQTT
- Module 3: IoT Devices
  - Sensors and Actuators
  - Connected Smart Objects
  - IP as the IoT Network Layer
  - Information Acquisition

# An IoT Application – Smart Fridge



# How Does the Fridge Do That?

- You are leaving the home (sense user)
- There's no milk in fridge (sense object)
- Use this information to make a decision (process)
- Inform user of decision (communicate)

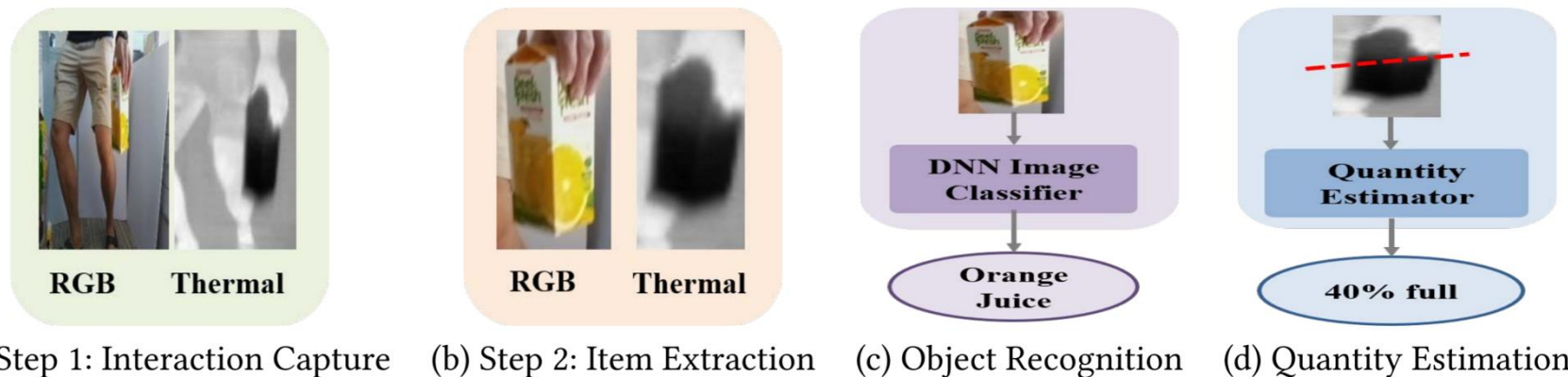
# How Does the Fridge Do That?

- **You are leaving the home (sense user)**
  - What type of sensor?
  - Distinguish between parent and child
  - Identify reason for leaving home
  - Identify other contexts (e.g., store hours)
- There's no milk in fridge (sense object)
- Use this information to make a decision (process)
- Inform user of decision (notify)

# How Does the Fridge Do That?

- You are leaving the home (sense user)
- **There's no milk in fridge (sense object)**
  - What type of sensor?
  - Is milk needed?
  - No milk or “little” milk? (prediction)
- Use this information to make a decision (process)
- Inform user of decision (notify)

# (Sensing often needs smart design)



**Figure 1: High-Level Steps in *SmrtFridge***

Sharma et al., SmrtFridge: IoT-based, user interaction-driven food item and quantity sensing, In the Proceedings of The 17<sup>th</sup> ACM Conference on Embedded Networked Sensor Systems (SenSys'19), Nov 2019, New York.

# How Does the Fridge Do That?

- You are leaving the home (sense user)
- There's no milk in fridge (sense object)
- **Use this information to make a decision (process)**
  - Where is processor?
  - What are the rules?
  - Fixed rules versus dynamic rules (learning)
- Inform user of decision (notify)

# How Does the Fridge Do That?

- You are leaving the home (sense user)
- There's no milk in fridge (sense object)
- Use this information to make a decision (process)
- **Inform user of decision (notify)**
  - How?
  - When?
  - Privacy?

# Needed Technologies

- Sensing technologies
  - Obtain data
- Connectivity technologies (key for IoT)
  - Exchange data
- AI technologies
  - Process data
- Cloud technologies
  - Backend for storage and compute

# Think More Broadly

- Individual reminder apps
  - Milk reminder app
  - Fruit reminder app
  - Xxx reminder app
- Or holistic smart home assistant?
  - Sounds like an open-set problem
  - Needs general AI?
  - What inputs?

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# What is IoT?

- The basic premise and goal of IoT is to  
    **“connect the unconnected.”**
- IoT is a technology transition in which devices will allow us to sense and control the physical world by making objects smarter and connecting them through an intelligent network
- Instead of viewing IoT as a single technology domain, **IoT is an umbrella** of various concepts, protocols, and technologies, all of which are at times somewhat dependent on a particular industry

# Augment Existing Things



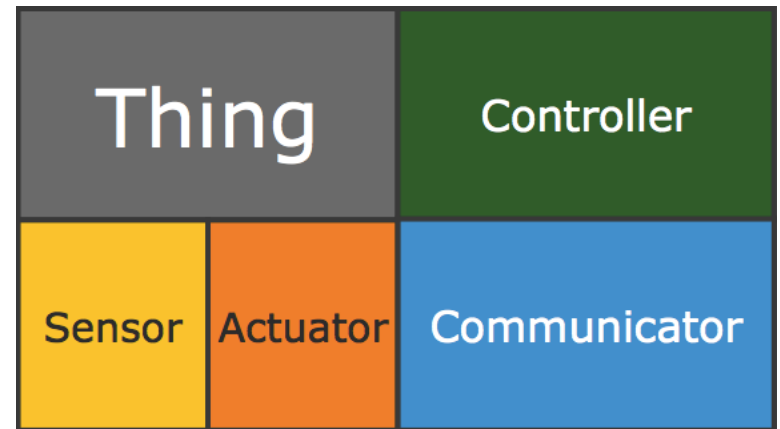
# Augmenting Life with New Things

- Smart city
- Smart car
- Smart health



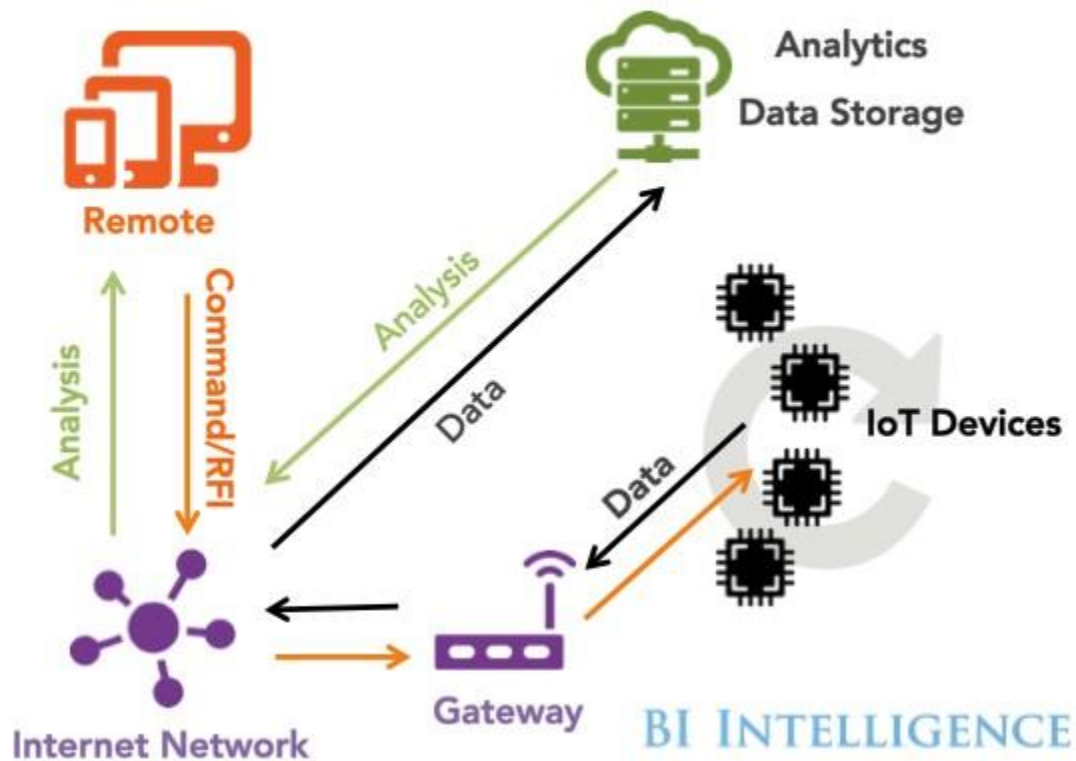
# IoT Device

Physical object (“thing”)  
+  
Controller (“brain”)  
+  
Sensors  
+  
Actuators  
+  
Networks (Internet)



# IoT System Domains

## The Internet of Things Ecosystem



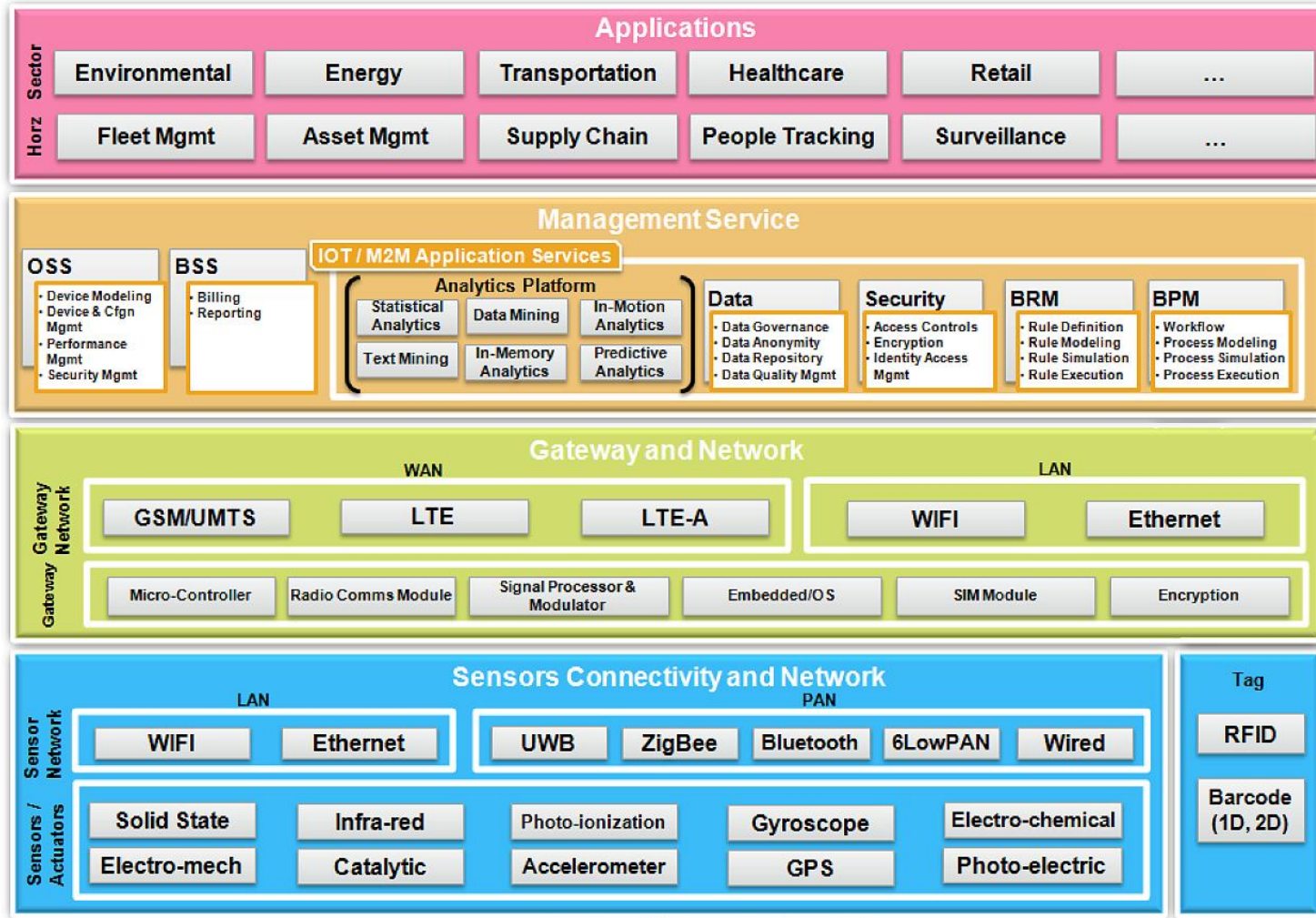
# IoT Architecture

**OSS:**  
Operational Support System

**BSS:**  
Business Support System

**BRM:**  
Business Rule Management

**BPM:**  
Business Process Management



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# IoT History

- First IoT device



1982

A modified Coke machine at CMU becomes the first internet connected appliance capable of reporting its inventory

# IoT History (cont'd)

- 1997, “The Internet of Things” is the seventh in the series of ITU Internet Reports originally launched in 1997 under the title “Challenges to the Network”.
- 1999, Auto-ID Center founded in MIT
- 2003, EPC Global founded in MIT
- 2005, Four important technologies of the internet of things was proposed in WSIS conference.
- 2008, First international conference of internet of things: The IoT 2008 was held at Zurich.

# IoT Growth

## THE INTERNET OF THINGS

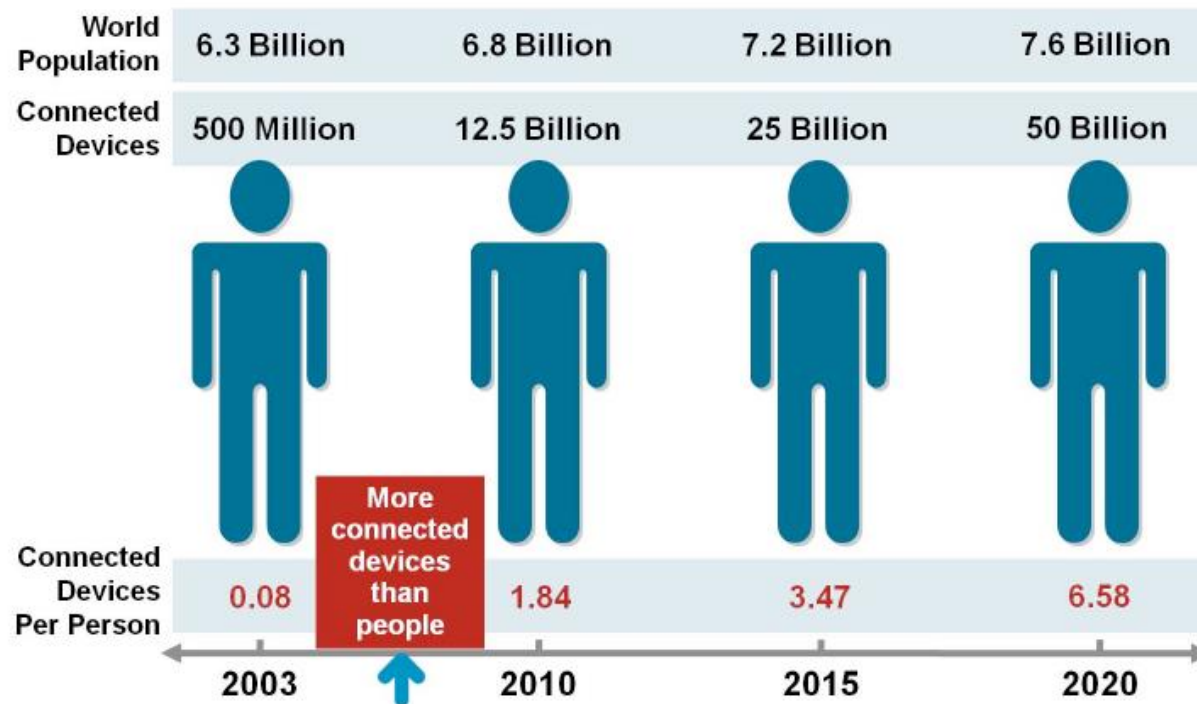
Connected devices (billions)



|                  | 15 billion | 28 billion | CAGR 2015–2021 |
|------------------|------------|------------|----------------|
| Cellular IoT     | 0.4        | 1.5        | 27%            |
| Non-cellular IoT | 4.2        | 14.2       | 22%            |
| PC/laptop/tablet | 1.7        | 1.8        | 1%             |
| Mobile phones    | 7.1        | 8.6        | 3%             |
| Fixed phones     | 1.3        | 1.4        | 0%             |

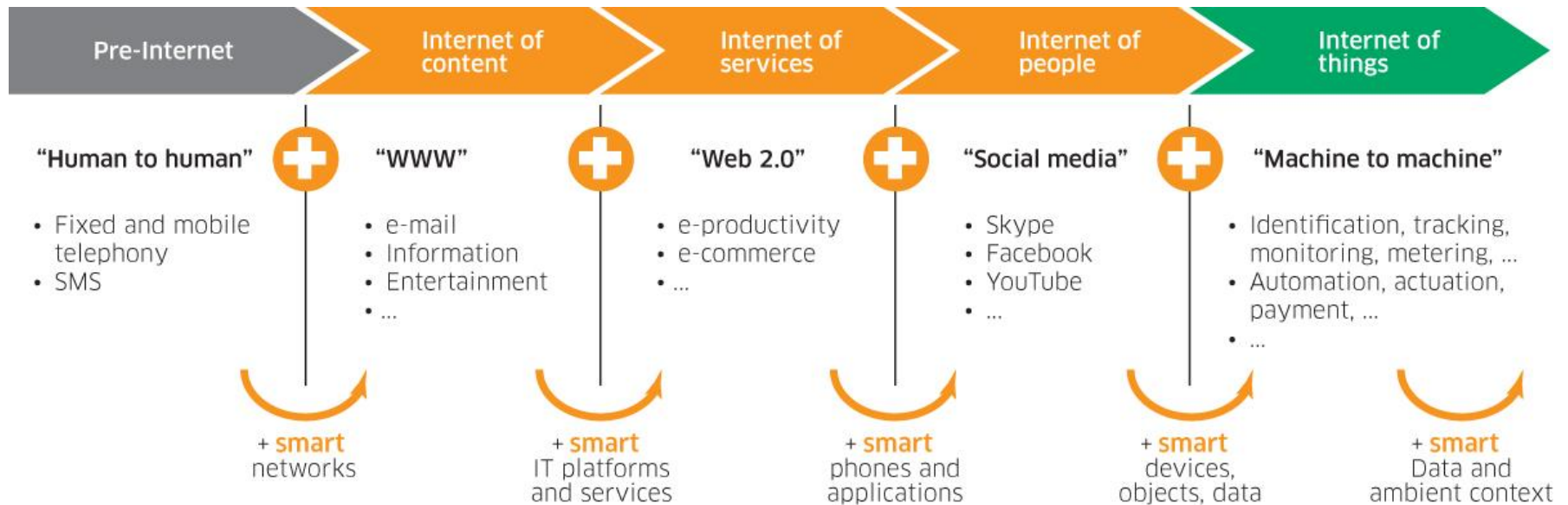
# IoT Growth (cont'd)

Figure 1. The Internet of Things Was “Born” Between 2008 and 2009



Source: Cisco IBSG, April 2011

# IoT Vision



# IoT Technology Trends

- **Miniaturisation of devices**
  - The size (and cost) of electronic components that are needed to support IoT capabilities such as sensing, tracking and control mechanisms, are the major factors in IoT applications
- **Increasing communication throughput and lower latency**
  - IoT relies on data communications and networking technologies to support “everything and everywhere and everyone” connectivity to occur
- **Real-time Analytics**
  - In today’s decision making process, the availability of real-time, accurate information is crucial.
  - With the growing volume of data from connected devices, social media platforms, etc, good decision making relies heavily on advances in analytic capability technologies, to bring out the intelligence in data.
- **Cloud Computing**
  - IoT connects billions of devices and sensors to create new and innovative applications.
  - To support these applications, a reliable, elastic and agile platform is essential and cloud computing is one of the enabling platforms to support IoT.

<https://www.imda.gov.sg/-/media/imda/files/industry-development/infrastructure/technology/internetofthings.pdf%3Fla%3Den>

# IoT Use Cases

- Autonomous and connected vehicles (Internet of Vehicles, IoV)
- Smart security for businesses, homes, and public safety
- Smart home
- Smart health
  - Smart watches, fitness trackers, and other wearables
- Industrial IoT (IIoT)
- Smart cities
  - Transportation, parking, and public space management
- Smart power grid
- Smart farm

<https://blog.westerndigital.com/top-9-iot-use-cases/>